DIGITAL CIRCUITS AND COMPUTER ENGINEERING, LAB1 REPORT

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Introduction

Lab 1: Lab Environment and Introduction to Digital Circuits
The learning outcomes of the labs are listed below:

- Use the breadboard and other devices in the lab environment and connect digital circuits with integrated circuits. Furthermore, connect power supply, switches (for input), and LEDs (for output).
- Verify the truth table for the basic logical operators: NOT, OR, AND, XOR, NAND, and NOR.
- Be able to verify truth tables for a digital circuit.
- Combine multiple individual gates into a digital circuit.

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1 Logic Gates

The main components that will be used in the labs are the logic gates. In this course, you will use fixed logic gates in the form of dual in-line package (DIP) integrated circuits (IC), see Figure 1.



Figure 1: Logic Gates

1.1 AND Gates

We created a very basic circuit with only one AND gate. We used the 7408 IC that contains 4 separate AND gates.

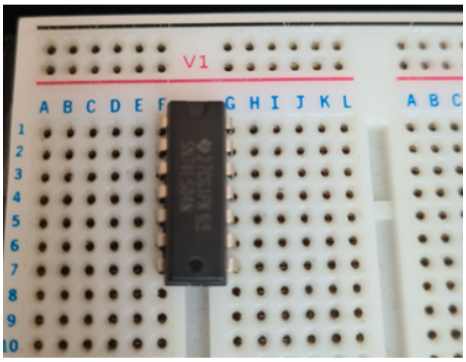


Figure 2: AND Gate

1.2 OR Gate

The IC that we have used is the 7432, which contains 4 OR gates, each with 2 input signals.

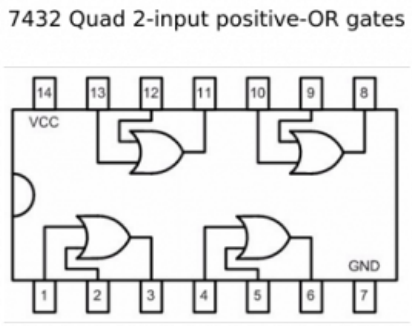


Figure 3: OR Gate

1.3 NOT Gate

The NOT gate we use is the 7404 IC, which contains 6 NOT gates.

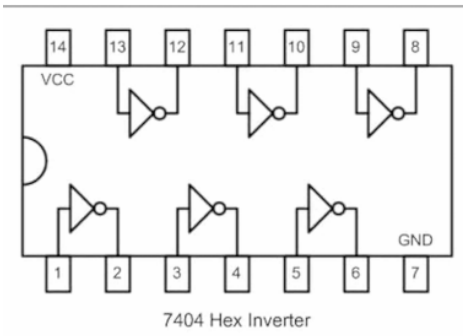


Figure 4: NOT Gate

1.4 Combinational Logic Gates

The first circuit consists of one single AND gate and one NOT gate (inverter). The output from the AND gate will be input to the NOT gate.



Figure 5: NAND Gate

NOR Gate

Take the input A and B from switches SW2 and SW3, and output X to LED 1. You need to first input A and B into NOT gates. Then take the outputs from the two NOT gates and input them to an OR gate.



Figure 6: NOR Gate Circuit

2 Lab Questions and Answers

2.1 Generate all possible input combinations of two binary variables A and B by flipping the switches SW1 and SW0. Observe the output of each combination (LED 0) and generate a truth table.

Answer 1:

Table 1: Truth Table for AND Logic

A	**B**	**Y = A · B**
0	0	0
0	1	0
1	0	0
1	1	1

2.2 Compare the truth table with the truth table of an AND gate presented in the coursebook.

Answer 2:

Table 2: Comparison of the AND Truth Tables

A	**B**	**Y = A · B**
0	0	0
0	1	0
1	0	0
1	1	1

A	**B**	**X**
0	0	0
0	1	0
1	0	0
1	1	1

2.3 Turn the power on and verify the truth table of the OR gate. Compare with the OR gate presented in the textbook.

Answer 3:

Table 3: Comparison for OR Gates

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	1

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Our lab work shows that both the tables in the course book and the results from our lab are the same. In the lab, we use IC 7432 that contains 4 OR gates, each with 2 input signals as shown in Figure 3.

2.4 Turn the power on and verify the truth table of the NOT gate. Compare with the NOT gate presented in the textbook.

Answer 4.1:

Table 4: Comparison of the NOT gate truth table

A	B
0	1
1	0

A	B
0	1
1	0

Both truth table is same.In the lab, we used IC 7404 that contains 6 NOT Gate.

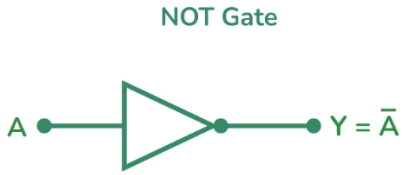


Figure 7: Not Gate

2.5 Turn the power on and generate the truth table. Verify that you get the truth table for a NAND-gate.

Answer 4.2:

Table 5: verification of truth table of NAND Gate

A	B	X
0	0	1
0	1	1
1	0	1
0	0	0

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

In the lab we used AND and NOT Gate to make a NAND Gate.

2.6 Turn the power on and generate the truth table. Compare the truth table with the truth table you got in Question 4.

Table 6: comparison of the NOR truth table with the Question 4.2

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

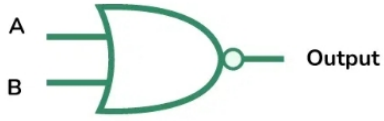


Figure 8: NOR GATES

Comparison A NAND gate and a NOR gate are both fundamental building blocks in digital electronics, known as universal gates because they can be used to construct any other logic gate. From their truth tables, we observe that a NAND gate outputs a low signal (0) only when both inputs are high (1), whereas a NOR gate outputs a high signal (1) only when both inputs are low (0). This

means NAND is the inverse of an AND gate, and NOR is the inverse of an OR gate. Despite their opposite behaviors, both gates are essential for designing complex logic circuits and are widely used in digital systems.

2.7 Turn the power on and generate the truth tables of the two circuits. Compare the truth tables.

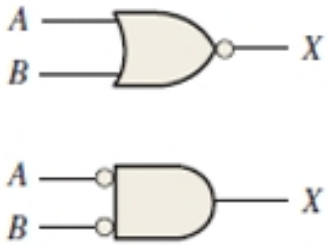


Figure 9: A NOR gate and a AND Gate with inverted input

Answer 6:

Table 7: Truth table for NOR and AND gate with inverted input

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

Table 8: NOR gate

A	B	X
1	1	1
1	0	0
0	1	0
0	1	0

Table 9: AND gate with inverted input

2.8 Turn the power on and generate the truth tables of the circuit. With your own words try to explain the logical behaviour of the circuit.

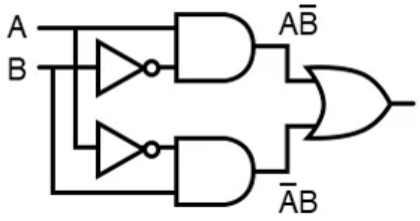


Figure 10: XOR

Answer:

Table 10: Table of XOR

A	**B**	**X**
0	0	0
1	0	1
0	1	1
1	1	0

An XOR gate outputs 1 only when the inputs are different—either A is 1 and B is 0, or vice versa. If both inputs are the same (both 0 or both 1), the output is 0. It's like a digital "inequality checker." This makes XOR useful in circuits that compare or toggle values.

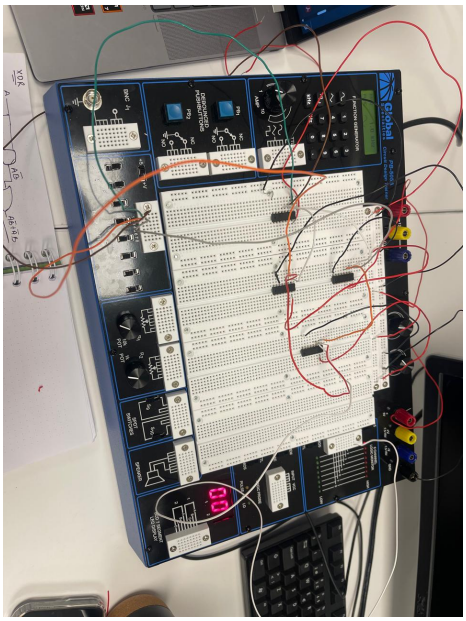


Figure 11: XOR circuit

2.9 What are the necessary steps that need to be carried out to build a circuit with fixed logic IC circuits given a circuit diagram? How can you test and verify that your circuit produce the correct output? How can you minimize the number of errors that you make during the lab work.

Answer : To build a circuit using fixed logic ICs from a given circuit diagram, the first thing to do is to study and familiarize yourself with the diagram thoroughly, noting the type of logic gates utilized and the connections between them. The second thing to do is to acquire all the required components, including the appropriate ICs, power supply, and wiring equipment. Mount the circuit on a breadboard or similar setup by carefully positioning ICs with regard to pin orientation and connecting inputs, outputs, and power according to the diagram. To test and confirm proper function of the circuit, implement all input combinations and cross-reference output observed with anticipated values from the truth table, using LEDs or measurement equipment such as multimeters to double-check for accuracy. Reducing errors throughout lab work is a matter of working systematically: checking each connection twice prior to energizing the circuit, having components clearly laid out, testing the circuit in small pieces whenever possible, and writing down each step so it is easy to spot and correct errors. Remaining organized and patient throughout will minimize

the likelihood of error and result in a successful build.

3 References

References

[1] CIRCUIT GLOBE,"AND gate,"[Online].Available: https://circuitglobe.com/and-gate.html. Accessed: Sep. 14, 2025.

[2] Greekforgeek, "OR,NOT, XOR, NAND"[Online].Availabe https://www.geeksforgeeks.org/digital-logic/universal–gates/ Accessed: sep.16,2025.